

Predicting 30-Day ICU Readmission in Coronary Heart Disease Patients Using Supervised Machine Learning Models: A Comparative Study of Logistic, Gradient-Boosted, and Neural-based Models

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Structured abstract

Objective:

This study aimed to develop and compare machine learning models for predicting 30-day ICU readmission among patients with coronary heart disease (CHD) using the MIMIC-IV clinical database. Three CHD subcohorts — Myocardial Infarction (MI), Percutaneous Coronary Intervention (PCI), and Coronary Artery Bypass Graft (CABG) — were defined according to PheKB guidelines. The objective was to evaluate how different algorithms and imbalance-handling techniques influence predictive performance.

Materials and Methods:

A merged CHD dataset of 15,445 ICU stays (across 11185 patients) with 53 features was constructed. Features included demographics, comorbidities, laboratory and vital-signs, and prior ICU admissions. We split the dataset by patient, race, and cohort into training, validation, and test sets. Three model families were implemented with multiple variants each for class imbalance handling:

1. Logistic Regression: LASSO and inverse class weighted.
2. Gradient Boosting (XGBoost): baseline, oversampled, class weighting.
3. Multilayer Perceptron (MLP): baseline/vanilla, oversampled, focal loss.

Result:

Among 3 MLPs, Focal Loss had the highest (among all models) test AUC-PR and AUC-ROC (0.194 and 0.742 resp.), showing the best discriminative ability but with poorly calibrated probabilities. SHAP-based feature importance on Focal Loss MLP reveals consistency of significant predictors of readmission to previous literature. After threshold tuning, Focal Loss MLP has higher recall than Vanilla MLP (0.549 vs 0.123) at the cost of precision (0.13 vs 0.185). Logistic regression exhibited moderate discrimination, with no sampling LASSO as the best (AUC-PR = 0.161 and AUC-ROC = 0.747). Oversampling and inverse class weighting improved recall (roughly 0.68) but slightly reduced discrimination (AUC-ROC roughly 0.73). XGBoost achieved stable performance across all variants, with the random-oversampling variant attaining the best AUC-PR = 0.178 and AUC-ROC = 0.722.

Conclusion:

All models demonstrated moderate ability to predict 30-day ICU readmission for CHD patients, with Focal Loss MLP showing the strongest overall discriminative performance. Gradient boosting exhibited

consistent discriminative performance across all variants. Future work could investigate more principled grid search approaches and domain-specific feature engineering to improve discriminative performance, followed by probability calibrations to increase medical usability (e.g. for clinicians or discharge planning).

Introduction & Prior Work

The project aims to use supervised machine learning on the MIMIC dataset to predict which patients with coronary heart disease (CHD) are at high risk of ICU readmission.

Chronic conditions affect 6 in 10 Australians, with CHD being a leading contributor to disease burden. It is responsible for about 305,000 disability-adjusted life years (DALYs) annually ([AIHW, 2024](#)). ICU readmissions worsen outcomes and place strain on hospital capacity. Prior Australian work examined building explainable models to predict readmission risk in chronic patients ([Koopman et al., 2020](#)); we adapt this approach to the ICU setting. Our goal is to identify high-risk CHD patients before discharge to enable timely intervention and reduce preventable readmissions.

Regarding CHD-related readmission prediction tasks, prior work (largely USA based) on modelling heart disease/heart failure readmissions on structured data revealed random forest and XGBoost (*eXtreme gradient boosting*) models to be the most commonly used ML models, as stated from a systematic review of 13 studies that built predictive models for 30 day heart failure readmissions by Son & Yu (2024). Logistic regression is a commonly used interpretable baseline ([Gao et al., 2023](#)), however one study on 30-day heart failure readmission prediction found the performance of Logistic Regression and Neural Network to be apparently comparable ([Allam et al., 2019](#)). Therefore, we consider 3 classes of models for evaluation with several variants corresponding to class imbalance handling strategies:

1. Logistic Regression: LASSO & LASSO resampling & inverse class weighted
2. Gradient Boosting/XGBoost: baseline, oversampled (random resampling), class weighting.
3. Multilayer Perceptron (MLP): baseline/vanilla, oversampled, focal loss.

Methods

Cohort Definition

Our study used the Medical Information Mart for Intensive Care (MIMIC-IV) database, a publicly available and de-identified resource containing detailed clinical information from the Beth Israel Deaconess Medical Center between 2008 and 2019 ([Johnson et al., 2023](#)). Each patient admission in MIMIC-IV is uniquely identified by `subject_id`, `hadm_id`, `stay_id`, and includes time-stamped records of vital signs, laboratory measurements, diagnoses, procedures, and outcomes.

To study coronary heart disease (CHD) and its relation to ICU readmission, we followed Phenotype KnowledgeBase (PheKB) algorithms ([Kirby et al., 2016](#)) to curate three CHD subcohorts:

1. Myocardial Infarction (MI)-identified through diagnostic codes. Categories for these cohorts include: acute-MI, post-MI complications, chronic Ischemic Heart Disease (IHD), and other types of acute IHD
2. Percutaneous Coronary Revascularization (PCI)-identified through procedural codes
3. Coronary Artery Bypass Graft (CABG)-identified through procedural codes

Patients were identified using ICD-9/10 diagnosis and procedure codes from the PheKB resource. However, phenotypes for acute MI patients (i.e. those belonging in the acute MI cohort with ICD-9 = ['%410'] or ICD-10 = ['I21%', 'I22%'], where '%' = any character) were further refined with elevated troponin ≥ 0.40 ng/mL to confirm acute myocardial injury ([Newby et al., 2012](#); [Thygesen et al., 2018](#)). We only rely on ICD-based phenotyping for other MI cohort types, primarily because they do not rely on acute biomarker events and are generally captured well via diagnosis codes.

The primary outcome was ICU readmission within 30 days of discharge, determined from sequential ICU stay timestamps. Records were linked across the patients, admissions, and icustays tables using `subject_id`, `hadm_id`, and `stay_id`.

Feature Definition and Engineering

After obtaining the labelled ICU stays, we extracted the following measurements:

Predictor variables were selected based on established clinical evidence and data availability within the MIMIC-IV database ([Newby et al., 2012](#); [Tang et al., 2019](#); [van Diepen et al., 2014](#); [Quan et al., 2005](#); [Vincent et al., 1996](#); [Caruana et al., 2015](#)).

The goal was to comprehensively represent each patient’s demographic profile, chronic health burden, acute physiological status, and care utilization patterns, all of which have been linked to 30-day ICU readmission risk in prior studies ([Tang et al., 2019](#); [van Diepen et al., 2014](#)).

Demographics included age, sex, ethnicity, and admission type (emergency, elective, urgent), extracted from the patients and admissions tables. Age and admission type have repeatedly demonstrated predictive value for readmission among cardiac and post-surgical ICU populations ([Tang et al., 2019](#); [Quan et al., 2005](#)).

Comorbidities were quantified using the Charlson Comorbidity Index (CCI) derived from ICD-9 and ICD-10 codes in `diagnoses_icd` ([Quan et al., 2005](#)). Although ICD 9 and 10 codes are technically generated at the end of hospital admissions, using CCIs as features does not significantly risk leakage as they identify chronic conditions that a patient already has at the beginning of admissions, rather than from complications arising during the stay ([Quan et al. 2005](#)). Specific comorbid conditions were also modeled as binary indicators, including hypertension, diabetes mellitus, chronic kidney disease, chronic obstructive pulmonary disease (COPD), heart failure, and malignancy. These features capture the long-term disease burden influencing recovery and stability after cardiac interventions.

Illness severity was characterized using composite ICU scoring systems — SOFA, SAPS II, and OASIS — available from `mimiciv_derived.sofa`, `sapsii`, and `oasis` tables ([Vincent et al., 1996](#)).

Vital signs (from `chartevents` and `mimiciv_derived.vitalsign`) encompassed both the mean and slope of last 24-hour change. Key variables included heart rate, systolic/diastolic blood pressure, mean arterial pressure, respiratory rate, oxygen saturation (SpO₂), and temperature. Trends and volatility in vital signs immediately prior to discharge have been shown to predict early readmission ([Tang et al., 2019](#)).

Laboratory results included troponin, creatinine, blood urea nitrogen (BUN), sodium, potassium, glucose, hemoglobin, white blood cell count, and lactate.

These markers assess myocardial injury, renal function, electrolyte balance, infection, and perfusion — domains closely tied to cardiac instability and readmission risk ([Newby et al., 2012](#); [Thygesen et al., 2018](#); [Caruana et al., 2015](#)).

Treatment and intervention variables were added to capture therapeutic intensity: number of administered medications, use of inotropes or vasopressors, mechanical ventilation, renal replacement therapy, and transfusion events.

Such indicators of care complexity are informative for estimating the likelihood of post-ICU complications.

ICU utilization and temporal factors included interval between previous discharges. **Longer LOS and multiple prior ICU stays** are known predictors of readmission ([van Diepen et al., 2014](#)).

Outcome Definition

The primary outcome was ICU readmission within 30 days of the previous ICU discharge, 30 days being a commonly adopted readmission time measure of hospital quality and patient recovery stability ([Tang et al., 2019](#); [Caruana et al., 2015](#)).

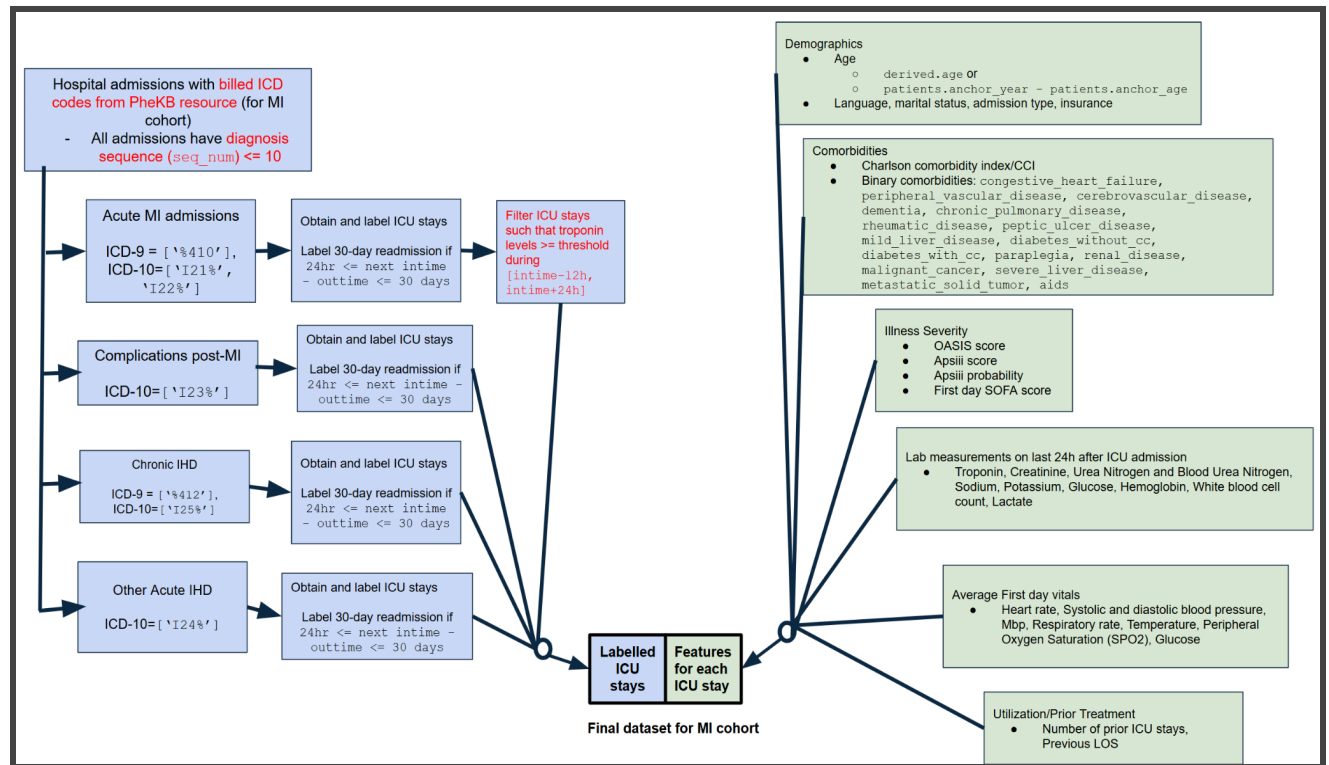
Each ICU stay was represented by unique identifiers (subject_id, hadm_id, stay_id), and the corresponding admission (intime) and discharge (outtime) timestamps were used to determine sequential stays.

For each patient, ICU stays were sorted chronologically by intime. If the next intime occurred within 30 days ($\leq 30 \times 24$ hours) after the preceding outtime, the earlier stay was labeled as readmitted = 1; otherwise, it was labeled readmitted = 0. This binary variable ($y \in \{0, 1\}$) was used as the target outcome for machine-learning prediction. The 30-day definition aligns with prior studies on readmission prediction and hospital performance benchmarking ([Tang et al., 2019](#); [van Diepen et al., 2014](#)).

Since many measurements (e.g. vital signs and laboratory results) are collected within 24 hours of an ICU stay, to minimize temporal leakage in the prediction task, we will consider features available **within this initial 24-hour window after ICU admission** as predictors available in prediction time. This implies that an extra condition is required to label readmission, namely that the next intime must occur at least 24 hours after the current outtime. Formally, the refined condition would be 'readmitted=1' if 'next intime - current outtime $\in [24 \text{ hours}, 30 \text{ days}]$ '. In other words, the prediction is made using data observed in '[intime, intime + 24h]'.

Dataset Description & Exploratory Data Analysis

Below is a graphical summary of our workflow for obtaining the MI cohort. All the other two cohorts have similar phenotyping workflow (except no lab measurements are used to refine the other two cohorts' phenotypes)



Quick data description. Before preprocessing, our combined dataset (across all 3 cohorts) consists of 15445 ICU stays (rows) across 11185 patients with 53 recorded features for each ICU stay, which consist of 24 non-binary numeric features, 20 binary features, 6 categorical features, and 3 identifiers. Out of the 15445 stays, 914 ICU readmissions were recorded (i.e. 5-6% readmission prevalence). 5430, 2349, and 7666 ICU stays come from the MI, PCI, and CABG cohort respectively.

Missing values. There are 3 features with at least 28% (>4000) missing values (troponin level, lactate level, and previous Length of Stay/LOS), and the rest have <8% (<1100) missing values. It turns out that there are >83% missing values for the previous LOS which are also *Missing at Random* (MAR), namely that if the number of prior ICU stays is 0 then the previous LOS would not be well-defined. This implies that conventional statistic-based imputation for previous LOS would be ill-defined due to MAR, so we dropped the previous LOS from any downstream analysis. Further, some of the features' missing values

(e.g. troponin & lactate level), seem to correlate with cohort placement. This is verified by performing a χ^2 -test of independence between the troponin level (and lactate) missingness indicator and the cohort label, which yielded a *Cramér's V* of > 0.4 , indicating evidence that the features' missingness may differ among the 3 cohorts. While this suggests cohort-specific numerical imputations, we did not implement this due to time constraints, however future work could explore this imputation method to minimize bias from imputation.

Correlation between features. Correlation analysis revealed no significant or near-perfect multicollinearity in the dataset; maximum observed pairwise Pearson correlation is 0.7.

Outliers. Univariate outlier analysis revealed two variables with rare (if not physiologically impossible) values: mean glucose of 999,999 (likely a placeholder) and white blood cell counts $> 400 (10^9 / \text{liter})$, so we exclude observations (3 in total) corresponding to these extreme values from modelling. Although Tukey boxplots for other variables show many data points falling beyond the $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ limit, we decided not to take further action to handle these supposed outliers due to insufficient clinical justification.

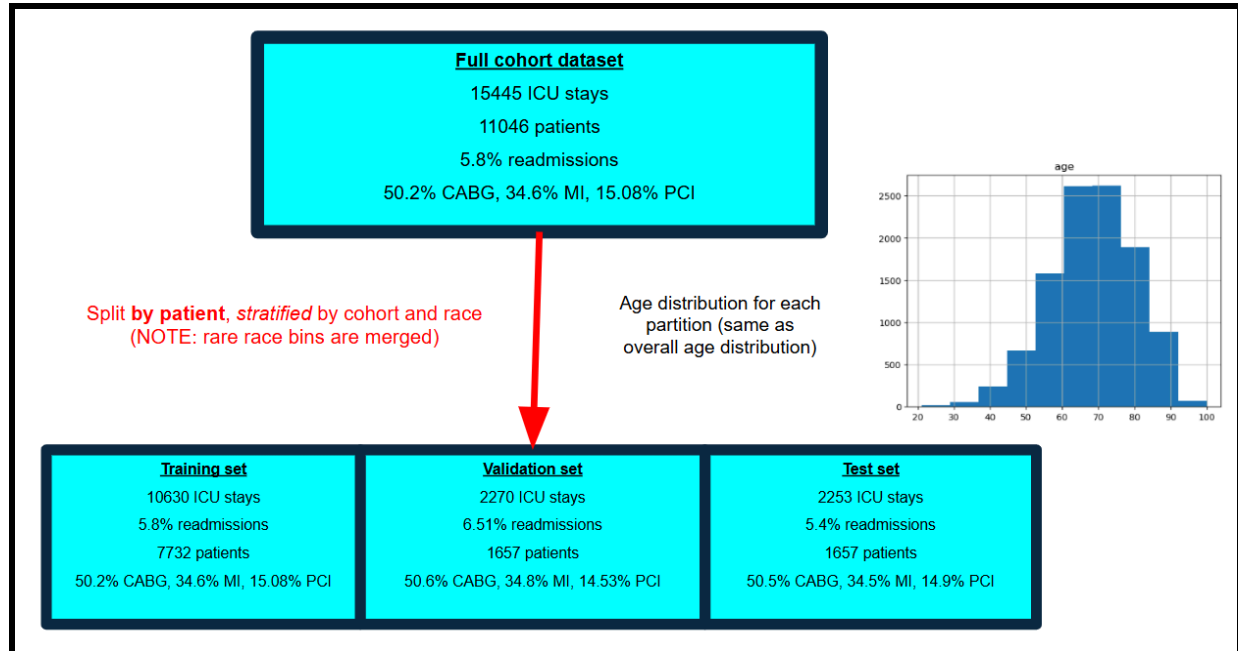
Preprocessing

To prepare our dataset for modelling, all continuous variables were standardized using z-score normalization to ensure comparability across measurement scales. Categorical variables were one-hot encoded, and missing values were imputed using the median for continuous data or the mode for categorical data, resulting in 119 numerical features in total. Preprocessing pipelines were identical across all models so that performance differences are expected to originate from the model itself.

Modelling Workflow

Dataset splitting

We split the dataset **by patient** (to prevent within-patient admission leakage) to training, validation, and test sets, stratified by cohort and race to preserve subgroup balance. Each model uses the **same** training, validation, and test sets. Below is a graphical diagram to illustrate the dataset partition:



Logistic Regression

We first develop an interpretable baseline model using logistic regression to establish a transparent reference for predicting ICU readmission. This baseline serves as a foundation for developing more advanced and interpretable models in subsequent experiments.

To establish baseline, two logistic regression models were implemented:

1. LASSO regression (with resampling and without resampling): A logistic regression model with L1 regularization (penalty='l1', solver='liblinear'), which promotes sparsity by shrinking less important coefficients toward zero (without resampling). A variant using Random Oversampling (ROS) was also applied to handle class imbalance by duplicating minority-class samples (with resampling).
2. Inverse class-weighted regression: trained with (class_weight='balanced'), which automatically adjusts sample weights based on class frequencies to reduce bias toward the majority class.

All models used the same hyperparameters (C=1.0, max_iter=2000).

Models were trained on the combined training and validation data, and an optimal decision threshold was determined on the validation set by maximising the F1-score across thresholds from 0 to 1.

Performance was assessed using ROC-AUC, PR-AUC, Accuracy, Precision, Recall, F1-score.

Gradient Boosting

We used the XGBoost classifier to address the challenge of class imbalance in our dataset. Initially, we trained candidate models without any resampling or class weighting to identify optimal hyperparameters. Based on this baseline configuration, we compared two modelling strategies:

1. Random Oversampling: we oversampled the training set by using RandomOverSampler from the imblearn.over_sampling package to balance classes.
2. Class Weighting: The model was trained on the original imbalanced dataset. We use the 'scale_pos_weight' parameter set to the ratio of negative to positive samples in the training set.

We used the grid search to tune the hyperparameters. The main selection criterion is average precision (PR-AUC) on the validation set. Because of the imbalance, we applied a threshold of 0.2 to convert predicted probabilities into binary labels for final evaluation.

Performance was assessed using Accuracy, ROC-AUC, PR-AUC, Precision and recall.

MLP

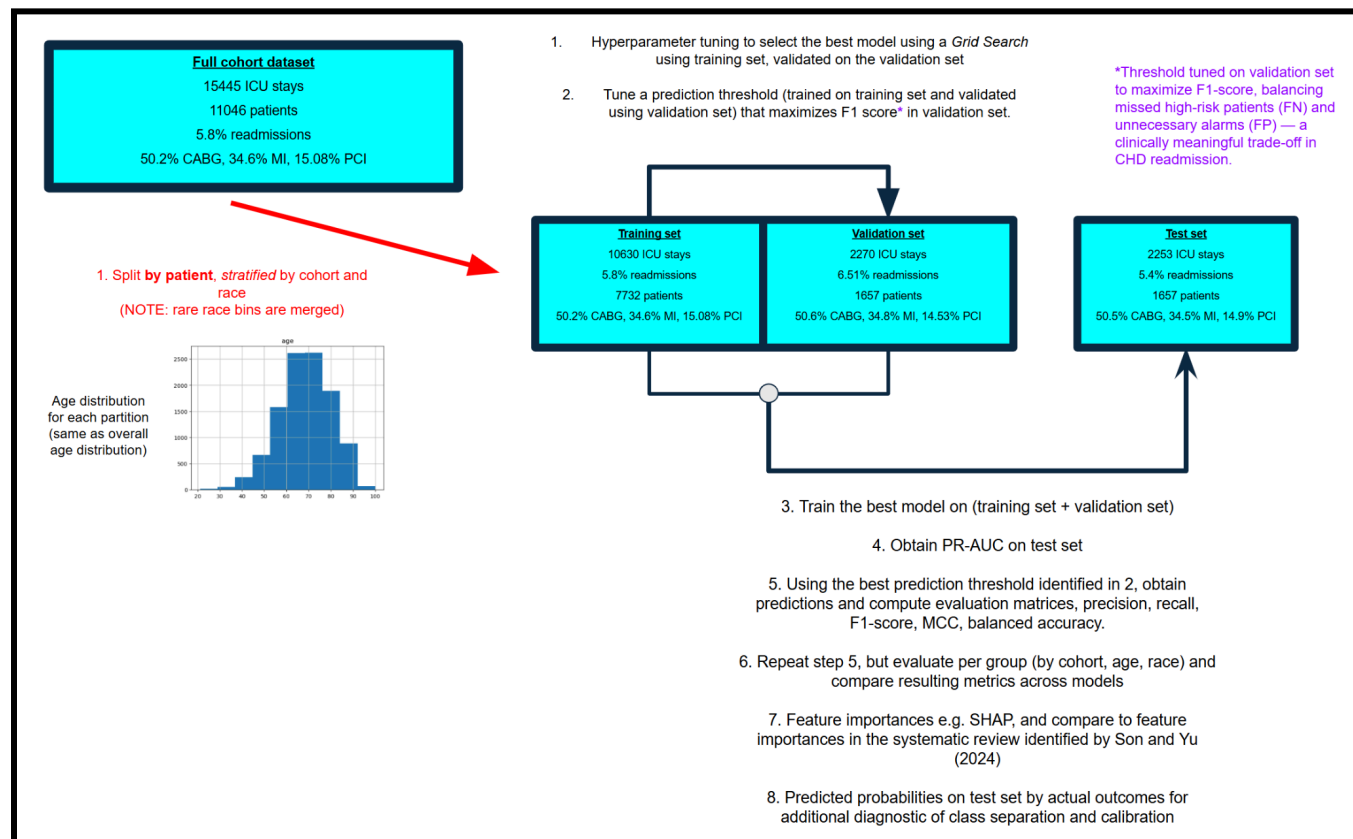
The figure below illustrates the entire workflow for training, validating, and evaluating MLP models. After splitting the dataset, hyperparameter tuning (via grid search) is performed on each of three types of Multi-Layer Perceptron (MLP) models that differ by class imbalance handling:

1. the “vanilla” MLP with binary cross entropy loss (no class imbalance handling),
2. the MLP trained on an oversampled training set (to balance positive and negative classes), and
3. the MLP with a focal binary cross-entropy loss as proposed by [Lin et al. \(2018\)](#), motivated by the possibility of oversampling to induce overfitting or cohort distributional distortion

(Note: for more information on the focal loss objective and the MLP grid search methodology, see the appendix)

The best weights during tuning are chosen to maximize the validation set’s AUC-PR (the area under the precision-recall curve) to address class imbalance. Following selection, models are refitted on the combined training + validation and evaluated on a held-out test set by AUC-PR and evaluation metrics from the model using the optimal threshold tuned in the validation set (Precision, Recall, F1 score,

Matthew's Correlation Coefficient/MCC, Balanced Accuracy, TN, FP, FN, TP). The workflow also includes group-level evaluation and SHAP-based post-hoc explainability. Predicted probability histograms were considered to diagnose class separation and calibration, **but not used to tune the model**.



(Note: for oversampled MLP, before hyperparameter tuning, the minority class samples (i.e. readmitted ICU stays) are resampled to achieve a 50-50 class balance overall in the training set. The class distribution of the validation set remains unchanged.)

Ethics Statement

This study utilized the MIMIC-IV database, a publicly available and fully de-identified critical care Laboratory for Computational Physiology. The database contains the anonymized clinical information collection from the Beth Israel Deaconess Medical Center between 2008 and 2019. All patient identifiers have been removed in accordance with the Health Insurance Portability and Accountability Act (HIPPA) standards, and access to the data required completion of the Collaborative Institutional Training Initiative (CITI) “**Data or Specimens Only Research**” course and acceptance of the MIMIC data use agreement.

Results

Logistic Regression

Below is the result from LASSO without resampling. The strong class imbalance is clearly reflected where the model predicts all samples as non-redmissions, indicating imbalance handling is necessary.

```
Best threshold: 0.101 F1: 0.257

=== LASSO (no resampling, tuned threshold) ===
ROC-AUC: 0.747344046895542
PR-AUC : 0.16134768149325118
Confusion Matrix:
[[2130   1]
 [ 122   0]]

              precision    recall  f1-score   support

     0       0.946        1.000        0.972        2131
     1       0.000        0.000        0.000         122

   accuracy          0.945
  macro avg       0.473        0.500        0.486        2253
 weighted avg     0.895        0.945        0.919        2253
```

Model	Accuracy	ROC-AUC	PR-AUC	Recall (y=1)	Precision (y=1)
LASSO (no resampling)	0.945	0.747	0.161	0.000	0.000
LASSO (resampling)	0.683	0.734	0.142	0.680	0.188
Inverse-weighted LR	0.688	0.738	0.143	0.689	0.193

Interpretations:

1. The base LASSO model failed to identify positive readmissions due to the strong class imbalance (~5% positive rate).

- Both oversampling (ROS) and class weighting improved recall from 0.00 to ~ 0.68 , confirming their effectiveness in handling imbalance.
- The similar performance between ROS and class-weighted LR suggests that class reweighting is a simpler yet effective alternative for logistic models on imbalanced medical data.

Gradient Boosting

```
param_grid = {
    "model__n_estimators": [500, 1000, 1500],
    "model__max_depth": [6, 8],
    "model__learning_rate": [0.01, 0.05],
    "model__subsample": [0.8, 0.9],
    "model__colsample_bytree": [0.8, 0.9],
    "model__reg_lambda": [1.0],
    "model__min_child_weight": [1],
}
```

We used the XGBoost model and did hyperparameter tuning with grid search. And, we chose the baseline model which has the highest PR-AUC on the validation set (0.1619, as shown below). Here is the test performance:

```
Best AUC-PR (val): 0.16190354586574174
===== 📊 Test Set Evaluation =====
Accuracy : 0.6924101198402131
ROC AUC  : 0.7237731842973743
AUC-PR   : 0.1615108602164842
```

	precision	recall	f1-score	support
0	0.971	0.696	0.811	2131
1	0.106	0.631	0.182	122
accuracy			0.692	2253
macro avg	0.538	0.664	0.496	2253
weighted avg	0.924	0.692	0.777	2253

The optimal parameter set is:

```
{'colsample_bytree': 0.9, 'learning_rate': 0.01, 'max_depth': 8, 'min_child_weight': 1, 'n_estimators': 500, 'reg_lambda': 1.0, 'subsample': 0.8}
```

Based on this, we have two strategies to better model:

- Random Oversampling: Oversampling the minority class in the training set
- Class Weighting: Using `scale_pos_weight` calculated from the class ratio

They were evaluated on the same test set, with a fixed threshold of 0.2.

Method	Accuracy	ROC-AUC	PR-AUC	Recall (Minority)	Precision (Minority)
Random Oversampling	0.7133	0.7223	0.1781	0.607	0.110
Class Weighting	0.8726	0.7118	0.1580	0.377	0.179

Key Observations:

1. The random oversampling model achieved the highest PR-AUC. It shows the best balance between precision and recall for the minority class.
2. The class-weighted model improved precision but at the cost of recall.

MLP

Below are results and figures from our tuning and test set evaluation using the 3 MLP variants.

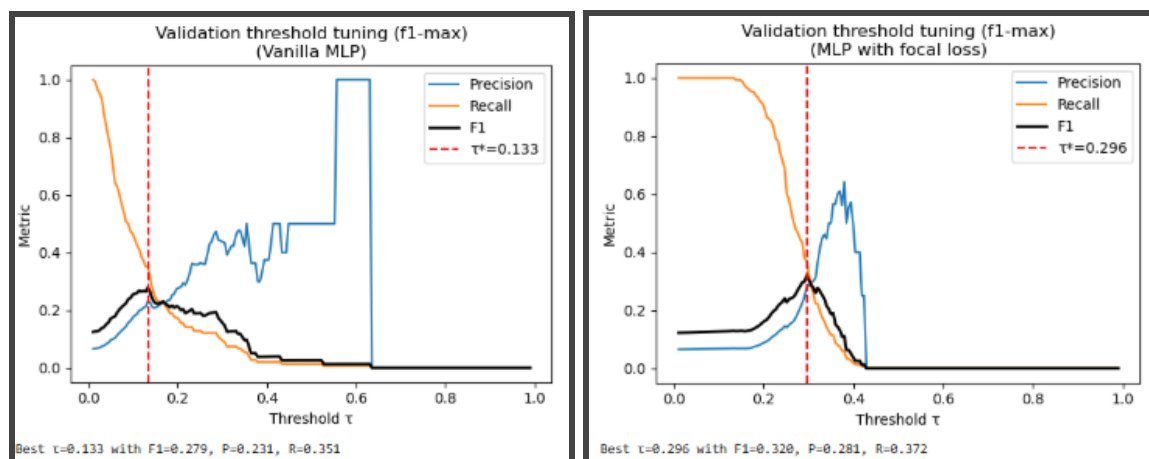
Best Models From Validation

MLP Type	Best Model Hyperparameters	Validation Set and Test Set AUC-PR <i>Best weights in training maximize validation AUC-PR. EarlyStopping is applied with patience=10. AUC-PR's "random baseline" is class prevalence, which is 0.065 for validation set and 0.054 for test set</i>
Vanilla MLP with Binary Cross-Entropy Loss (i.e. class imbalance not addressed)	3 hidden layers with neurons = [256, 128, 64] Activation function = ReLU BatchNormalization = applied Dropout = 0.3 ElasticNet regularization (both L1 and L2 penalty 1e-6) Batch size = 450	Validation set AUC-PR = 0.180 Test set AUC-PR = 0.111

MLP with Minority Class Resampling	3 hidden layers with neurons = [256, 128, 64] Activation function = Leaky ReLU BatchNormalization = applied Dropout = 0.2 ElasticNet regularization (both L1 and L2 penalty 1e-4) Batch size = 450	Validation set AUC-PR = 0.206 Test set AUC-PR = 0.144
MLP with Focal Loss	3 hidden layers with neurons = [128, 64, 32] Activation function = Leaky ReLU BatchNormalization = applied Dropout = 0.2 ElasticNet regularization (L1 penalty = 1e-3, L2 penalty = 1e-4) Batch size = 256 Focal $\alpha = 0.5$, Focal $\gamma = 2$	Validation set AUC-PR = 0.244 Test set AUC-PR = 0.194

Threshold Tuning (Vanilla MLP & MLP with Focal Loss)

Tuning is done on the validation set (not on test set). Optimal F1-maximizing thresholds for vanilla MLP and Focal Loss MLP are 0.133 and 0.296 respectively.



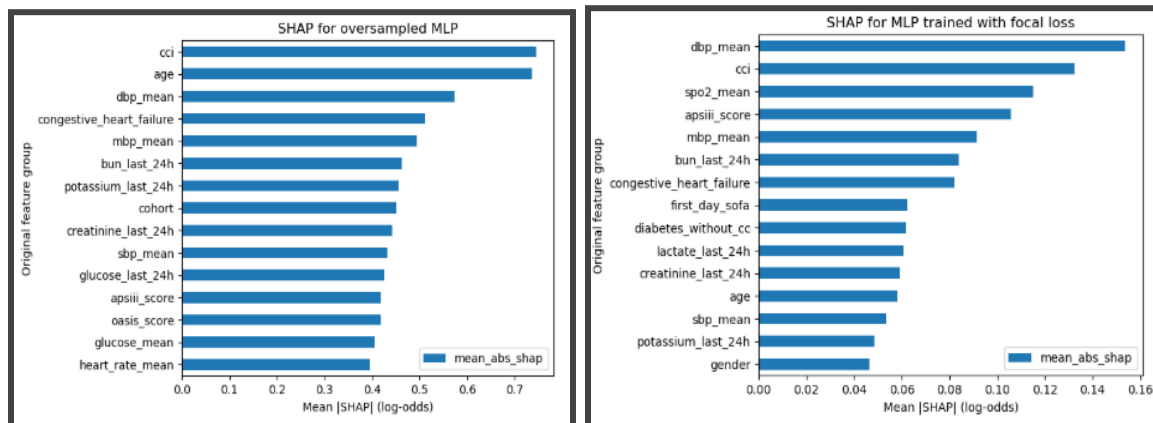
Model Evaluation using Optimal Threshold

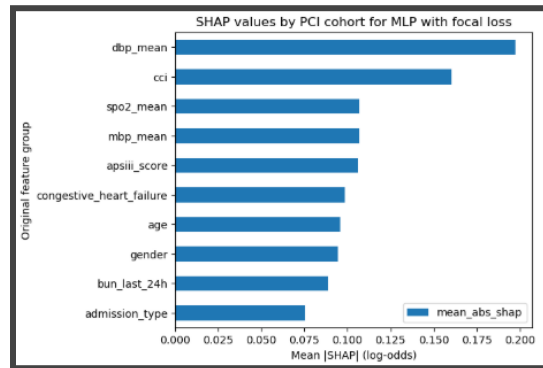
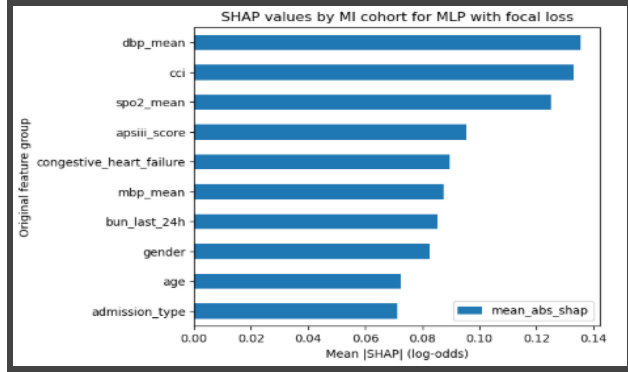
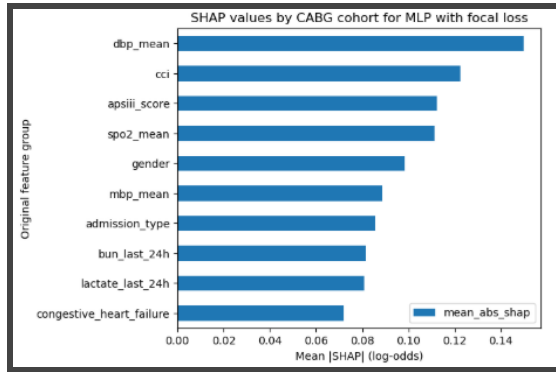
MLP Type	Precision	Recall	F1	MCC	Balanced Accuracy	TN	FP	FN	TP
Vanilla	0.185	0.123	0.148	0.112	0.546	2065	66	107	15
Focal Loss	0.130	0.549	0.210	0.182	0.669	1682	449	55	67

Evaluation by Cohort (Focal Loss MLP with Optimal Threshold)

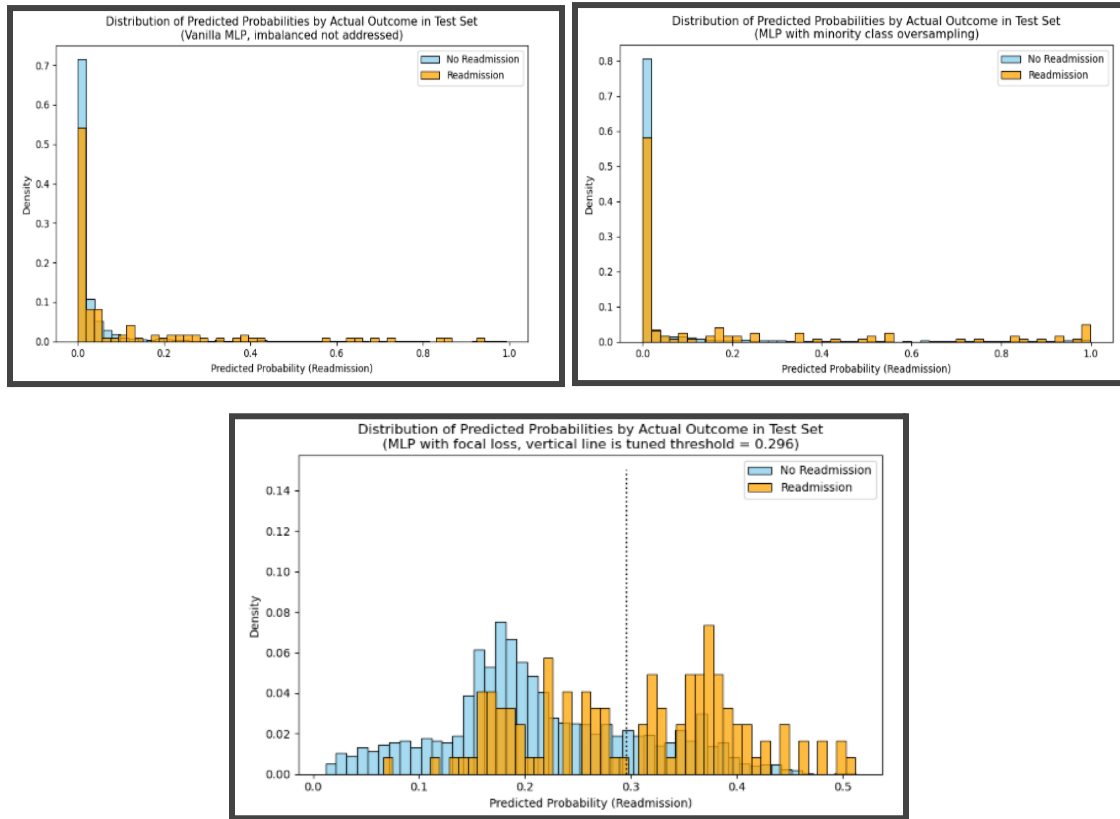
Evaluation by Cohort														
Group	N	Positives	Prevalence	Precision	Recall	F1	BalancedAccuracy	MCC	TN	FP	FN	TP		
0	CABG	1138	51	0.044815	0.190476	0.392157	0.256410	0.656980	0.224455	1002	85	31	20	
1	MI	779	58	0.074454	0.119205	0.620690	0.200000	0.625879	0.135644	455	266	22	36	
2	PCI	336	13	0.038690	0.100917	0.846154	0.180328	0.771374	0.223586	225	98	2	11	
----- Evaluation by (Binned) Age -----														
Group	N	Positives	Prevalence	Precision	Recall	F1	BalancedAccuracy	MCC	TN	FP	FN	TP		
2	60-69	668	40	0.059880	0.176991	0.500000	0.261438	0.675955	0.222719	535	93	20	20	
3	70-79	646	29	0.044892	0.126761	0.620690	0.210526	0.709859	0.209864	493	124	11	18	
1	50-59	356	19	0.053371	0.121622	0.473684	0.193548	0.640403	0.155546	272	65	10	9	
0	0-49	148	6	0.040541	0.111111	0.500000	0.181818	0.665493	0.169026	118	24	3	3	
4	80+	435	28	0.064368	0.106250	0.607143	0.180851	0.627896	0.130178	264	143	11	17	
----- Evaluation by (Collapsed) Race -----														
Group	N	Positives	Prevalence	Precision	Recall	F1	BalancedAccuracy	MCC	TN	FP	FN	TP		
1	BLACK	121	8	0.066116	0.296296	1.000000	0.457143	0.915929	0.496464	94	19	0	8	
6	WHITE	1564	84	0.053708	0.126984	0.571429	0.207792	0.674228	0.183497	1150	330	36	48	
0	ASIAN	49	4	0.081633	0.142857	0.250000	0.181818	0.558333	0.091287	39	6	3	1	
5	UNKNOWN/OTHER	448	24	0.053571	0.114943	0.416667	0.180180	0.617531	0.133801	347	77	14	10	
2	HISPANIC	62	2	0.032258	0.000000	0.000000	0.000000	0.366667	-0.107676	44	16	2	0	
3	PACIFIC	4	0	0.000000	0.000000	0.000000	0.000000	0.500000	NaN	4	0	0	0	
4	PORTUGUESE	5	0	0.000000	0.000000	0.000000	0.000000	0.400000	NaN	4	1	0	0	

SHAP (Oversampled MLP & MLP with Focal Loss)





Test Set Predicted Probabilities by Test Set Actual Outcome



Test AUCs Across All Models

The table below summarizes test set evaluation across all models.

Candidate Models	Test AUC-PR	Test AUC-ROC
Logistic Regression (LASSO; no resampling)	0.161	0.747
Logistic Regression (LASSO; resampling)	0.142	0.734
Logistic Regression (Inverse Class Weighted)	0.143	0.738

Baseline XGBoost (train as is)	0.1615	0.7238
Random Oversampling XGBoost	0.1781	0.7223
Class Weighted XGBoost	0.1580	0.7118
Vanilla MLP	0.111	0.68
Oversampled MLP	0.144	0.692
Focal Loss MLP	0.194	0.747

Discussion & Conclusion

Comparing performance across all models. To summarize, among all models, the Focal Loss MLP has both the highest AUC-PR (0.194) and the highest AUC-ROC (0.747). Among logistic regression variants, the LASSO model without resampling obtained the highest AUC-ROC (0.747) and a comparable AUC-PR (0.161), while both oversampled and inverse-weighted variants produced slightly lower AUC-ROC values (0.734 and 0.738) with similar AUC-PRs (roughly over 0.14). Discrimination abilities are relatively stable among XGBoost variants, but the random-oversampling XGBoost variant achieves the best AUCs (AUC-PR = 0.178, AUC-ROC = 0.722). The ranking of models ordered by decreasing AUC-PR are: (1) Focal-loss MLP (2) Oversampled XGBoost (3) Logistic Regression with LASSO & no resampling, while ranking by decreasing AUC-ROC include (1) Logistic Regression with LASSO & no resampling $\approx$ Focal-loss MLP (2) Inverse Weighted Logistic Regression (3) Baseline XGBoost. The first 2 models maximizing AUC-PR employed class imbalance handling, underscoring the importance of addressing imbalanced class distributions in our CHD dataset. After threshold tuning, focal-loss MLP showed a substantial gain in minority-class recall (0.549) compared to the Vanilla MLP (0.123) at the cost of precision (0.13 vs 0.185), which could reflect the expected trade-off of the focal-loss objective. Inverse weighted-logistic regression achieved recall = 0.689 and precision = 0.193, and the oversampled XGBoost achieved recall = 0.607 and precision 0.110 under the 0.2 threshold. Traditional logistic linear models maintain slightly higher ROC discrimination but the focal-loss MLP offers the best precision-recall balance for this highly imbalanced CHD readmission task.

Summarizing MLP Results in deeper detail. The best MLPs across all 3 types have the same depth with 3 hidden layers and ElasticNet regularized weights, but the best focal-loss based MLP has slightly lower width trained on a smaller batch size (see “Results” → “Best Model from Validation” section). The optimal focal loss hyperparameters are $\alpha = 0.5$ and $\gamma = 2$. The highest validation set AUC-PR is attained from the focal loss MLP (0.244), followed by MLP with resampling (0.206) and vanilla MLP (0.180). After model selection, the chosen MLPs were re-trained on the combined training + validation data before being evaluated on an unseen test set. The ranking of the models’ AUC-PRs remain consistent: 0.194, 0.144, and 0.111 for focal loss MLP, oversampled MLP and vanilla MLP respectively.

Model calibration is complementary to overall discriminative ability to assess its usability, which primarily includes threshold calibration and probability calibration.

- **Threshold calibration.** Threshold calibration is largely context-specific. For example, [Golas et al. \(2018\)](#) developed a deep learning model for 30-day heart-failure readmission using a threshold that maximizes the “projected cost-saving to the hospital”. This motivates the need to tune a prediction threshold on the validation set to optimize a case-specific benefit; here we choose a threshold that maximizes the F1-score, addressing the need for timely intervention (i.e. reducing False Negatives) as well as hospitals’ finite resources and capacity (i.e. reducing False Positives). The optimal thresholds for the vanilla and focal loss MLPs are 0.133 and 0.296 respectively (see “Threshold Tuning” section); oversampled MLP’s threshold is very similar to vanilla MLP. Evaluation metrics are computed for vanilla MLP and focal loss MLP using their optimal thresholds (see “Model Evaluation using Optimal Threshold” section).
 - Moreover, to investigate subgroup performance differences, performances of models using the optimal threshold were marginally stratified by cohort (CABG, MI, PCI), age bands, and race groups (see “Evaluation by Cohort (Focal Loss MLP with Optimal Threshold)” section). Group-based evaluations indicate that the 60-69 age group has the highest F1-score, however false positives are concentrated among older patients (70-80 years). The highest F1-score by race is highest among Black patients, followed by White and Asian groups.
- **Probability calibration.** To address probability calibration (how well do predicted probabilities match true outcomes), we consider the predicted probabilities for the test set across all three MLP models by their actual outcomes (see the “Test Set Predicted Probabilities by Test Set Actual

Outcome” section). As expected, the predicted probabilities for the vanilla MLP are extremely skewed, even with oversampling minority classes. Although the evaluation metrics remained low for the threshold-calibrated MLP models (see “Model Evaluation using Optimal Threshold” section), the focal-loss MLP provides a cleaner separation of predicted probabilities. However, predicted probabilities in the test set for the threshold-calibrated focal loss MLP sits on the 0 - 0.5 range, indicating poor probability calibration. Later sections will discuss the implication of probability calibration.

To assess post-hoc explainability, a SHAP (*SHapley Additive exPlanations*) analysis was conducted with 200 samples from the test set as *background samples*. Across the oversampled and focal loss MLP variants as well as cohort-specific SHAP summaries indicate identical rankings of the most influential predictors, giving confidence to the robustness of MLP across different imbalance handling strategies (see “SHAP (Oversampled MLP & MLP with Focal Loss)” section). Top ~10 influential predictors include comorbidities, diastolic and mean blood pressure, congestive heart failure flag, BUN/creatinine, severity scores, admission type, and cohort flag. These align with the systematic review of 13 studies of 30-day heart failure readmission prediction tasks from [Yu & Son \(2024\)](#), namely the majority of (9/13) their reviewed studies found vital signs, medical history (including past history of MI and PCI) and comorbidities to be significant predictors for 30-day heart failure readmission. Such consistency gives confidence that the model learns “on the right track”.

More potential in improving discriminative ability for MLP. The focal loss MLP’s slightly superior ability for separation in their predicted probabilities likely means that it has captured more meaningful structure in our CHD readmission dataset than the other two counterparts. As mentioned, although evaluation metrics (e.g. precision and recall) are relatively low, these could still inform future modelling and feature designs. For instance, instead of the current *ad-hoc*, selective grid search, one could consider more principled grid search procedures (e.g. Bayesian optimization). Further, concentration of misclassifications in older patients and MI cases motivates additional manual feature engineering e.g. learned latent embeddings of trajectory-based features and interaction terms between comorbidities and vital signs.

What’s next after maximizing discriminative ability for our CHD readmission problem? High discrimination alone does not guarantee clinical usefulness as predicted probabilities may be poorly calibrated ([Alba et al. 2017](#)). Furthermore, a meta-analysis by [Van Grootven et al. \(2021\)](#) identified a major gap in models for identifying unplanned heart-disease related hospital admissions: they provide

good discriminative ability (e.g. high AUCs) with poor probability calibration. In the context of this project, for example, our focal loss MLP could have achieved a high AUC-PR yet predicted probabilities severely underestimate actual outcomes. Once discriminative capacity is maximized (e.g. building from focal loss MLP’s promising discriminative ability), future studies could investigate calibrating probabilities to additionally make them clinically interpretable. This allows threshold selection to meaningfully reflect risk tolerance, complementing the model’s discriminative ability to convince clinicians of the model’s usefulness to make readmission decisions. Furthermore, since cohort placement is among the influential predictors identified by SHAP in our MLP models, one could consider a mixed MLP model by cohort.

To sum up, accurately identifying high readmission-risk CHD patients before ICU discharge remains critical for optimizing hospital resources and mitigating preventable negative outcomes. Despite our candidate models’ average-to-below-par performance, the focal loss MLP’s modest improvement in discriminative gain could still be translated to operational and clinical gains if integrated with principled modelling, domain-guided feature engineering, and better calibration.

Future Work

We summarize future directions as follows:

1. Our setup assumes that ICU stays are independent, even within patients. A natural extension may involve modelling the *sequence* of a patient’s prior ICU admissions by treating each stay as a time step. For example, a Recurrent Neural Network (RNN) could take the last k ICU stays as input features and predict whether the most recent stay will be followed by another readmission within 30 days.
2. In real life, clinicians might be more interested in predicted probabilities (e.g. pre vs post-test probabilities) as opposed to a binary “flag readmission vs no-flag”. In addition to deploying reliable discriminative classification models, future work could consider calibrating predicted probabilities to produce a clinically interpretable threshold.
3. Consider, say, cohort-specific or age-specific thresholds instead of a global threshold for all predictions.
4. Consider developing a mixed classification model that accounts for cohort-level differences.
5. Consider within-cohort imputation strategies to better preserve group patterns.

6. Use more exhaustive or principled grid search procedures instead of *ad-hoc* cherry picking combinations.
7. One might want to consider resampling at the patient rather than ICU-stay level to increase the interpretability of cohort balancing.
8. The current troponin-based filtering for acute MI patients used a fixed universal threshold as a proxy for the 99th percentile upper reference limit (URL). However, in practice, troponin assays differ widely in sensitivity. Future work could incorporate assay-specific thresholds by empirically computing the 99th percentile for each laboratory item.
9. Finally, although our model was evaluated on an unseen test set as a proxy for external validation, the study remains limited by the single-center nature of MIMIC-IV, which may constrain the generalizability of the findings to other populations. Future work could incorporate multi-center data to enable retraining and fine-tuning to capture institution-specific variations.

Appendix

Focal loss. The sample-specific focal loss proposed by [Lin et al. \(2018\)](#) takes on the following form:

$$FL(p_t) = \alpha_t (1 - p_t)^\gamma \log(p_t)$$

Where $p_t = \hat{p}$, the sample's predicted probability, if the true label is 1 and $p_t = 1 - \hat{p}$ otherwise. The focal α_t hyperparameter is α for positive classes ($1 - \alpha$ otherwise) which serves as inter-class weighting. The focal γ controls how much 'easily classified' examples are downweighted.

Hyperparameter combinations used for model selection in MLPs.

Grid search for all three types of MLPs were performed greedily in the interest of time (i.e. a "two-step" method: start with a modest-to-small but overdispersed grid, then 'cherry pick' hyperparameter combinations 'near' the best combination to create a narrower grid). Here I only show the final grids (with the possible combinations being the *cartesian product* of the values in the python dictionary) used to obtain the models for analysis (left to right $\rightarrow$ vanilla MLP, oversampled MLP, focal loss MLP).

```

manual_grid_vanilla_1 = {
  # build_mlp knobs
  "neurons": [[128,64],[256,128,64]],
  "activation_name": ["relu", "leaky_relu"],
  "bn": [True],
  "dropout": [0.2, 0.3],
  "optimizer": ["adam"],
  "learning_rate": [3e-4],
  "reg_type": ["l1_l2", "none"],
  "reg_l1": [1e-6],
  "reg_l2": [1e-6],

  # training loop
  "batch_size": [250, 450],
  "epochs": [500]
}

manual_grid_4 = {
  # build_mlp knobs
  "neurons": [[128,64],[256,128,64]],
  "activation_name": ["relu", "leaky_relu"],
  "bn": [True],
  "dropout": [0.2, 0.3],
  "optimizer": ["adam"],
  "learning_rate": [3e-4],
  "reg_type": ["l2", "l1_l2"],
  "reg_l1": [1e-6, 1e-4],
  "reg_l2": [1e-6, 1e-4],

  # training loop
  "batch_size": [450],
  "epochs": [500]
}

focal_grid_2 = {
  "neurons": [[128,64,32], [256,128,64], [256,128,64,32]],
  "activation_name": ["leaky_relu"],
  "bn": [True],
  "dropout": [0.2],
  "optimizer": ["adam", "adamw"],
  "learning_rate": [3e-4],
  "weight_decay": [1e-4], # for adamw
  "reg_type": ["l1_l2"],
  "reg_l1": [1e-4, 1e-3],
  "reg_l2": [1e-4, 1e-3],

  # focal settings
  "loss_name": ["focal"],
  "focal_alpha": [0.5],
  "focal_gamma": [0.5, 1.0, 2.0],

  # training loop
  "batch_size": [128, 256],
  "epochs": [300],
}

```

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CRediT - Contributor Role Taxonomy

Contributor Roles (13)	Keith	Jane	Louis	Kai
Conceptualization	✓			
Data curation	✓	✓		✓
Formal Analysis	✓	✓		✓
Funding Acquisition (NOT RELEVANT)				
Investigation	✓			
Methodology	✓	✓	✓	✓
Project Administration	✓			
Resources (bigquery, jupyter notebook)	✓	✓	✓	✓
Software (python etc)	✓	✓	✓	✓
Supervision	✓			
Validation	✓	✓	✓	✓
Visualisation	✓	✓		✓
Writing - original draft			✓	
Writing - review & editing	✓		✓	